Cheng (Frank) Gao, PhD

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Summary

- **Biotech & AI investment professional** adept at communicating complex scientific concepts with experience independently conducting scientific, clinical, commercial, and valuation diligence, with a focus on **immuno-oncology and antibody–drug conjugates (ADCs)**
- **Interdisciplinary scientist** with **5+ years** developing machine learning solutions in single-cell genomics and protein design; builds **agentic investment research infrastructure** for sourcing and diligence

Work Experience

DIMENSION NYC, NY
Research Associate (Member of Investment Committee) *Feb 2026 – Present*

- Independently authored **10+ investment memos** integrating scientific diligence, competitive landscapes, commercial assessments, and valuation, with therapeutics work focused on **immuno-oncology and ADCs**
- Evaluated **clinical efficacy, safety, PK/PD, and biomarker strategies**, assessing patient selection, response durability to inform investment recommendations
- Built **asset-specific Monte Carlo models** of therapeutic sales, risk-adjusted valuation, financing, dilution, and investment returns, incorporating disease-specific eligible patient populations, patient access, competition, and modality-specific costs
- Identified **200+ companies** and conducted founder outreach for **50+ opportunities** across therapeutics (75%) and AI infrastructure (25%)
- Built an **agentic therapeutics diligence workflow used firm-wide** for first-pass asset analysis; developed RAG-based competitive intelligence and voice AI tools for live discussions
- Authored in-depth blog posts on **autoresearch for biology** and **targeting the surfaceome**

AbbVie North Chicago, IL
Intern, Single-Cell Machine Learning *June 2025 – Aug 2025*

- Independently conceived and implemented a novel **cell type annotation method** using a variational autoencoder and uncertainty quantification; achieved **over 95% accuracy** on two internal datasets comprising millions of cells; remains in internal use
- Assessed **cell type-specific delivery efficacy for two AAV assets** through transduction-efficiency analysis and presented findings to senior leadership; delivered a comparative deck evaluating technical claims and approaches across **five mRNA design companies**

Flagship Pioneering Boston, MA
Fellow, AI/ML *June 2024 – Aug 2024*

- Evaluated **technical feasibility of AI-enabled biotech ventures**, including contrastive learning to infer spatial omics from H&E histology; delivered **three white papers** and three company-wide presentations
- Built a **transformer model for molecular generation**, owning model architecture, data-splitting strategy, and initial training

Moderna Boston, MA
Co-op, Data Science and Artificial Intelligence *Jan 2024 – June 2024*

- Developed and deployed an **ensemble deep learning model** for chromatography analysis in JAX on AWS EC2; improved peak-identification accuracy from **70% to over 93%** and reduced inference time from **5 minutes to 3 seconds** per trace; remains in use by formulation and QC teams

8VC SF, CA
Bio-IT Fellow *April 2025 – June 2025*

- Developed and presented a **virtual-cell investment thesis** informed by literature review and interviews with **20+ experts across 10+ organizations**; recommended against pursuing the opportunity given low translation potential and technical noise in Perturb-seq and chemotranscriptomic data

ARCH Venture Partners Chicago, IL
Intern *Jan 2025 – April 2025*

- Screened **50 life sciences companies** focused on oncology and computational drug design; conducted **one in-depth diligence assessment** of novel immuno-oncology targets and patient-stratification methods

Personal Angel Investing

- Made **one personal investment** in a therapeutics company developing a potential first-in-class asset for anorexia; support founders through institutional investor introductions and drug development decision

Education

PhD, University of Chicago — Chicago, IL
Bioinformatics & Computational Chemistry — *Sep 2019 – Dec 2025*

- **Thesis**: ***Deciphering High-Dimensional Biological Data—From Delineating Transcriptional Dynamics to Unraveling Protein-Protein Interaction Landscapes***

B.S., University of California, Berkeley — Berkeley, CA
Chemistry (High Honors) — *Aug 2015 – May 2019*

Research Experience

University of Chicago — Chicago, IL
Graduate Researcher, Theoretical Chemistry & Bioinformatics — *Mar 2020 – Dec 2025*

- Conceived and implemented **TopicVelo**, a Python method for inferring cell-state transitions from scRNA-seq data; recovered known and identified new biology underlying inflammatory responses in a **mouse psoriasis model**, resulting in a **first-author PNAS publication**
- Developed a **contrastive learning and protein language model** approach in PyTorch to assess protein–protein interactions from **130K phage selection assays**; predictions were **validated in binding assays**, resulting in a co-first-author manuscript as computational lead
- Performed rigorous **genomic** analysis that revealed dynamic heterogeneity of macrophage states within **tumor microenvironment** in a mouse colorectal cancer to inform target identification
- Led foundational research applying **physics-informed AI** to infer spatiotemporal dynamics; authored the successful proposal for the lab's **first NIH R01 grant**

Investment & Technical Skills

Biotech Diligence: Target biology, preclinical translation, clinical efficacy/safety, PK/PD, trial design assessment, biomarkers, patient stratification, competitive landscapes
Investment Analysis: Monte Carlo simulation, risk-adjusted valuation, patient-based market sizing, financing/dilution, investment returns, sourcing and founder outreach
Computational Biology & AI: Single-cell/spatial transcriptomics, protein language models, AlphaFold3, RDKit, LAMMPS, generative models, contrastive learning, variational inference, agentic workflows, RAG
Tools: Python (PyTorch, JAX, Pyro, NumPy, Pandas, SciPy), R, SQL, C++, Bash, Git
Languages: Chinese and English (native fluency)

First & Co-First Author Publications

1. S. S. Lu[†], M. J. Styles[†], **CFG**[†], A. Nandy, C. Basile, J. A. Pixley, S. Tao, A. R. Dinner, S. Vaikuntanathan, and B. C. Dickinson[*], 'Mapping the topology of protein-protein interaction fitness landscapes' (In Review at *Nature*).
2. **CFG**, S. Vaikuntanathan[*], and S. J. Riesenfeld[*], 'Dissection and Integration of Bursty Transcriptional Dynamics for Complex Systems', **PNAS**, vol. 121, no. 18, p. e2306901121, 2024.

Awards

Polsky Commercialization Fellowship (2024): Awarded by the Polsky Center of Innovation, this award is given annually to ~ 5 students to support commercialization of a research project from a UChicago lab
Seymour Goodman Graduate Fellowship (2024): Awarded by the Department of Chemistry at UChicago, this award is given annually to 1-2 student(s) for excellence in research
Wayne C. Booth Graduate Student Prize for Excellence in Teaching (2021): Awarded by the University of Chicago, this top teaching prize is given ~ 5 students every year
Eckhardt Fellowship (2019): Awarded by the University of Chicago, this award is a 5-year scholarship for the top 10 percent of entering PhD students with the highest potential in research
High Honors in General Scholarship and Major (2019): Awarded by UC Berkeley for being top 10 percent in cumulative GPA among all graduates in the College of Chemistry with an honors thesis